

DML Commands using Clause Operators and functions in Queries

(Aim) to implement DML Commands using clauses, Operators and functions in Queries

Data manipulation Language (DML):

The DML is used to retrieve, insert and modify database information. these Commands will be used by all database users during the routine operation of the database

DML Commands:

1. Insert into this is used to add records into a relation

Syntax: Insert into <table name> (field1, field2, ..., fieldn)
Values (data1, data2, ..., data n);

Example: SQL > Insert into Customer Values (238, 'Ramu',

'Chennai', '986264090');

SQL > Insert into Customer Values (409, 'Rocky', 'Vizag', '844118092');

SQL > Insert into Customer Values (112, 'Virat', 'Hyderabad', '704986929');

After inserting:

Customer ID	name	address	ph. no
238	Ramu	Chennai	986264090
409	Rocky	Vizag	844118092
112	Virat	Hyderabad	704986929

2. Update Set. Where

this is used to update the contents of a record in a relation.

Syntax: SQL > update relation name Set Field name = data,
field name2 = data; where field name = data;

Example: SQL > update Customer Set name = 'kumar' where
Customer ID = 409.

Customer ID	name	address	ph. no
238	Ramu	Chennai	986264090
409	kumar	Vizag	844118092
112	Virat	Hyderabad	704986929

3 Delete -from

This is used to delete all the records of a relation but it will retain the structure of that relation.

a) Delete -from: This is used to delete all the records of relation.

Syntax: SQL > Delete from table-name;

Example: SQL > Delete from Customer;

After deleting:

Customer-ID	name	address	ph.no

b) Delete -from. where: This is used to delete a selected record from a relation.

Syntax: SQL > Delete from relation-name where Condition;

Example: SQL > Delete from Customer where name = 'Ravi';

After deleting:

Customer-ID	name	address	ph.no.
409	Kumar	Vizag	844118092
112	Virat	Hyderabad	704986929

Truncate

This Command will remove the data permanently. But structure will not be removed.

Syntax: Truncate Table <table-name>

Example: Truncate Table Customer;

After truncate:

Customer-ID	name	address	ph.no

Queries

1. Retrieve a member name starts with letter 'v';

Query: Select name from bank-account where name like 'v%';

Result =

Name
Nijay
Vikram
Virat

2. List of Accounts where Balance Between 10000 and 20000.

Query: Select * from Bank account where balance between 10000 and 20000.

Output

Name	Acc number	Balance	Category
Vijay	23415	10000	Savings
Vikram	7890	9000	Savings

3. Finding records who has minimum Balance

Query: Select * min(balance) - from Bank account;

Output: min(Balance)
10000

4. Finding records who has Balance $\geq$ 20000;

Query: Select * from Bank account where balance $\geq$ 20000

Output:

Name	Acc Num	Balance	Category
Vikram	7890	20000	Savings
Virat	4567	35000	Savings
Aakash	8987	50000	RD

5. Distinct

Query: Select distinct Category from Bank account;

Output: Category:

Savings
Salary
RD

6. Union

Query: Select name from Cuthery listen Select name from Bank account;

Output: Name

Rocky
Virat
Vijay
Vikram
Aakash

VELTECH	
EX No	108125
PERFORMANCE	5.7
RESULT	5
VIVA VOCE	5
PROJECT	5
THAT DATE	15

Result: the implementation of DML Commands using Clauses, operators and function in Queries executed successfully.

Aggregate functions

Aim: To Start & implement aggregate functions (Count).

Sum(), Avg(), Min(), max()) on a sample database

Aggregate function

(are mostly used with grouped by to group the rows

→ Count()

→ Sum()

→ Avg()

→ Min()

→ Max()

CREATE TABLE STUDENT 2

ROLL NO INT PRIMARY KEY

NAME VARCHAR (50).

AGE INT,

DEPT ID INT,

MARKS INT);

INSERT INTO STUDENT 2 VALUES.

(1, 'Arjun', 20, 101, 85)

(2, 'Sneha', 21, 101, 90)

(3, 'Ravi', 19, 102, 95)

(4, 'Priya', 22, 102, 95)

(5, 'Kiran', 20, 101, 60)

(6, 'Anita', 23, 103, 88)

SELECT * FROM STUDENT 2

	ROLL NO.	NAME	AGE	DEPT-ID	MARKS
1	1	Arjun	20	101	85
2	2	Sneha	21	101	90
3	3	Ravi	19	102	95
4	4	Priya	22	102	95
5	5	Kiran	20	101	60
6	6	Anita	23	103	88

SELECT DEPT_ID, AVG(MARKS) AS MARKS FROM STUDENTS
GROUPED BY DEPT_ID;

	DEPT_ID	AVG MARKS
1	101	78
2	102	82
3	103	88

SELECT DEPT_ID, MAX(MARKS) AS TOP MARKS FROM
STUDENT
GROUP BY DEPT_ID;

	DEPT_ID	TOP MARKS
1	101	90
2	102	95
3	103	88

SELECT DEPT_ID, MIN(MARKS) AS LEAST MARK FROM
STUDENT 2
GROUP BY DEPT_ID

	DEPT_ID	LEAST MARK
1	101	60
2	102	76
3	103	88

SELECT DEPT_ID, COUNT(*) AS STU, COUNT FROM STUDENT 2
GROUP BY DEPT_ID

	DEPT_ID	STU. COUNT	VELTECH	
			EX No.	3.2
			PERFORMANCE (S)	5
			RESULT AND ANALYSIS (1)	5
			VIVA VOCE (1)	3
			RECORD (4)	1
			TOTAL (11)	13
			SIGN WITH DATE	

Result:

Implementation of all aggregate function has been performed successfully on table.

Output :-

	DEPT. ID	STU. COUNT
1	101	3
2	102	2
3	103	1

